The new Lancet Commission identifies 17 catastrophic threats to human survival between 2025 and 2100 and warns decision-makers to take action.

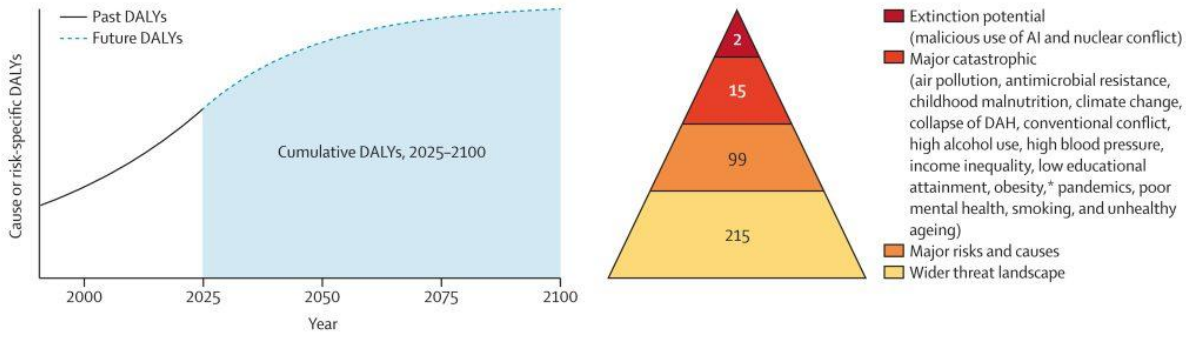
Global experts spanning health, economics, security, governance and technology have informed a new risk analysis in *The Lancet Commission on 21st Century Global Threats to Health*. Examining hundreds of possible threats to humanity covering 204 countries, authors identified 17 threats that pose the greatest danger to human health and survival between 2025 and 2100.

Experts also warned that, if allowed to continue, the 17 catastrophic threats could account for tens of billions of years of lost health. Every country faces significant risk, but compounding inequalities will see resource-limited and marginalised communities face the most health harms.

While health was of the greatest concern, these threats also pose dangers for global prosperity, security and stability. However, authors also found that governments, funders and industry still have a realistic route to act through policy change, funding decisions and innovation.

'The striking finding is not only the scale of these threats, but that humanity still has the agency to act,' says Commission co-chair Professor Christopher JL Murray. 'Across all 17 catastrophic threats, the decisions made by leaders today will shape the health and wellbeing of the next generation. But delaying action will only serve to shift and increase the costs into the future.'

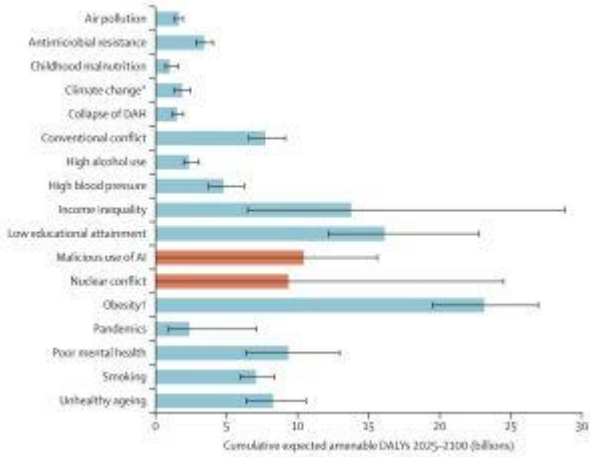
Alongside identifying each threat, the Commission modelled plausible best-case and worst-case future scenarios to allow leaders in health, finance, defence, climate and security sectors to address risks. The gap between each scenario was measured in amenable disability-adjusted life years (DALYs), or 'the number of years in good health that are lost due to sickness, disability or premature mortality that could be avoided with targeted action'.



A graph showing the potential number of good health years lost and the danger-level of causes threatening humanity in the next 75 years. Credit: Lancet Commission.

AI and nuclear conflict identified as extinction-level threats

Although the Lancet Commission notes that the chances of malicious use of AI and nuclear conflict remain low, they can still be identified as extinction-level threats due to potential impact and probability. Action to prevent nuclear conflict could avert an estimated 133 million deaths and 9.35 billion DALYs by 2100, while action to reduce the risk of malicious AI could avert an estimated 10.4 billion DALYs.



The 17 catastrophic threats to human health and survival. Credit: Lancet Commission.

Further assessment shows that climate change is already a catastrophic and urgent threat to humanity. The Commission finds that limiting global warming to around 1.8°C (rather than letting it reach 4.4°C) could prevent 1.89 billion years of healthy life from being lost to causes like extreme heat and air pollution.

Other key findings determine educational attainment, income inequality and obesity to be some of the greatest threats to human health and survival. Sub-Saharan Africa is forecast to bear around 70% of the amenable health burden linked to a collapse in development assistance for health.

Dr. Murray adds: ‘The estimates highlight how much health loss could be avoided in each area. Some are already imposing substantial health burdens today, while others carry a real danger if they materialise. Crucially, all 17 threats are interconnected, which is why they must be addressed together, not in isolation.’

Build, back and track: a global call to action

The Lancet Commission also published a common framework for leaders and decision-makers to inform action before harm becomes irreversible. For many of the catastrophic threats, solutions such as medicines, health technologies and proven public policies already exist to be implemented. For others, further targeted investment and innovation are required.

The Commission’s proposed framework follows a three-part agenda:

1. **Build readiness** – create and fund threat-ready health systems with dedicated teams for foresight, prevention, preparedness and protection across a wide range of risks. These capabilities should be embedded in how health systems are funded and delivered, supported by durable financing rather than ad-hoc crisis responses, and designed to reach lower-resource settings and marginalised communities, not just wealthier populations.
2. **Back breakthroughs** – increase investment in innovation where existing tools are insufficient, including new vaccines, antibiotics, obesity treatments and climate technologies, alongside appropriate governance and equitable access.
3. **Track threats** – reassess threats regularly and independently so that evidence keeps pace with emerging and changing risks and reaches decision-makers in finance and security as well as health, including the creation of the *Lancet Monitor of 21st Century Global Threats to Health* to provide an institutionalised mechanism for tracking emerging risks.

'Recent global events show us just how quickly stability can give way to crisis, and how costly it is to respond only after disaster strikes,' notes Commission co-chair, Dr. Kanem. 'The Commission makes a compelling case for reshaping the future by embedding foresight into decision-making. Governments and communities must act decisively starting today, to protect health and build a fairer, more peaceful and more secure future for the coming generations.'